Superposition and interference

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add signed displacements and distinguish interference from lost energy.

SUCCESS CRITERIA - I can...

1. Add wave displacements at the same place and time.
2. Recognize constructive and destructive interference.
3. Explain that pulses continue after crossing in a linear medium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “superposition” and “interference” to explain your answer. Sentence starters are optional.

superposition: adding displacements or fields at the same place and time

interference: the pattern produced by overlapping waves

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

An upward rope pulse is **3 cm** high before reaching an ideal free end. Is the reflected displacement upright or inverted? **Sketch it**.

Word help: ideal = using stated simplifying conditions

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $y_{resultant} = y_1 + y_2$.
Add signed displacements at the same place and time. Local cancellation does not destroy all wave energy.

Worked reasoning: At one point, two pulses have displacements **+3 cm** and **-1 cm**. Superposition gives **+2 cm** there. **Add signed displacements**, not their absolute values.

Your turn: Draw the two component displacements and the resultant at that point.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two upward pulses overlap. **Sketch the combined displacement**. Then **sketch equal upward**/downward pulses overlapping.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

Two rope pulses overlap. At point A their displacements are **+3 cm** and **-1 cm**; at point B they are **+2 cm** and **-2 cm**.
1. **Add signed displacements** at each point.
2. **Draw the two contributions** and each resultant.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Two rope pulses momentarily cancel. A student says the pulses are gone forever. **What happens** after they pass in an ideal linear medium?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A system receives **60 J** heat and does **40 J** work on its surroundings. Find ΔU with $\Delta U = Q - W_{by}$.
Label energy entering, leaving and stored.

Word help: surroundings = everything outside the chosen system

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

1. **Add displacements -4 mm** and **+7 mm**.
2. Name whether they reinforce or partly cancel.

Sentence starter: I notice … . This supports … because … .

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Construct three different pairs of displacements with resultant zero. Can the waves still move after that instant?

Sentence starter: Both … . They differ in … because … .

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3